

Three Amazing Things You Can Do With Raspberry Pi



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I keep trying new things with Raspberry Pi, or Raspi, to automate things around. Raspi never fails to surprise me with its ability to implement such complicated systems without any hassle. This article covers the implementation of cloud server, BitTorrent server and RasPlex media centre.

I have been thinking of making my own cloud server where I could keep confidential files and access these whenever needed, from any place, as I do not trust the various free cloud based services available in the market. After many failed at-

tempts, I found success in building such a server, ownCloud with Raspi. ownCloud logo is shown in Fig. 1.

Similarly, BitTorrent server was on my wishlist for a long time as I download a lot. Raspi showed me the straightforward path to achieve my goal.

I had been using Raspbmc media centre connected to an LED TV for quite some time now. But when someone suggested to me to try Rasp Plex media centre, I did so and was very impressed with its performance.

I find the three projects mentioned in this article to be quite exciting. Given next are the detailed steps for all three projects. Try these out, and you will find out how easy it is to build such systems with Raspi.

Your private cloud server

Everything is on cloud these days. People like the ease with which they can be at sync using one of the free cloud services such as Google and Dropbox. But, do you trust these servers for your confidential data? A lot of articles on the Internet explain how these free servers are actually not so free. Your most private data is stored on a server, over which you have no direct control.

This definitely creates requirement for a cloud where you can store data, on your very own hardware, with direct control. ownCloud7 is an interesting software that can help you imple-

ment this. It comes with various services like CalDAV, WebDAV and CardDAV that let you manage applications like calendar, music, contacts, files and much more.

Let us get started. Install everything on Raspi to make your own cloud server, which you can control as you like.

Prerequisites for the same are:

1. Ready-to-boot Raspi with Raspbian Wheezy OS
2. Internet connectivity for downloading and installing various software

First, boot Raspi. You can either install all software remotely or directly on Raspi; saving it remotely saves you from connecting the monitor and other unnecessary clutter to Raspi.

Run the terminal and follow the steps below to set up Raspi, to make it ready for installing all software:

1. Expand the root file system to have enough space for the cloud. You can do it by typing `sudo raspi-config` in the terminal and then selecting `expand_rootfs` as shown in Fig. 2.
2. Select `change_locale` and change it to `en_US.UTF8`
3. Select `memory_split` and allocate 16M to video graphics.

Once the above set-up is done, update packages using commands given below:

```
$ sudo apt-get update (ignore the $ sign and copy the commands only)
$ sudo apt-get upgrade
```

Create the required user and group with commands given below:

```
$ sudo groupadd www-data
$ sudo usermod -s -G www-data www-data
```

(*Your installation might already have the group and user created earlier)



Fig. 1: ownCloud logo

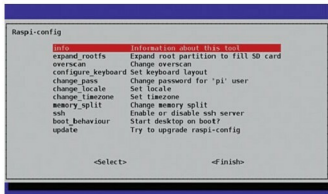


Fig. 2: Raspi-config window